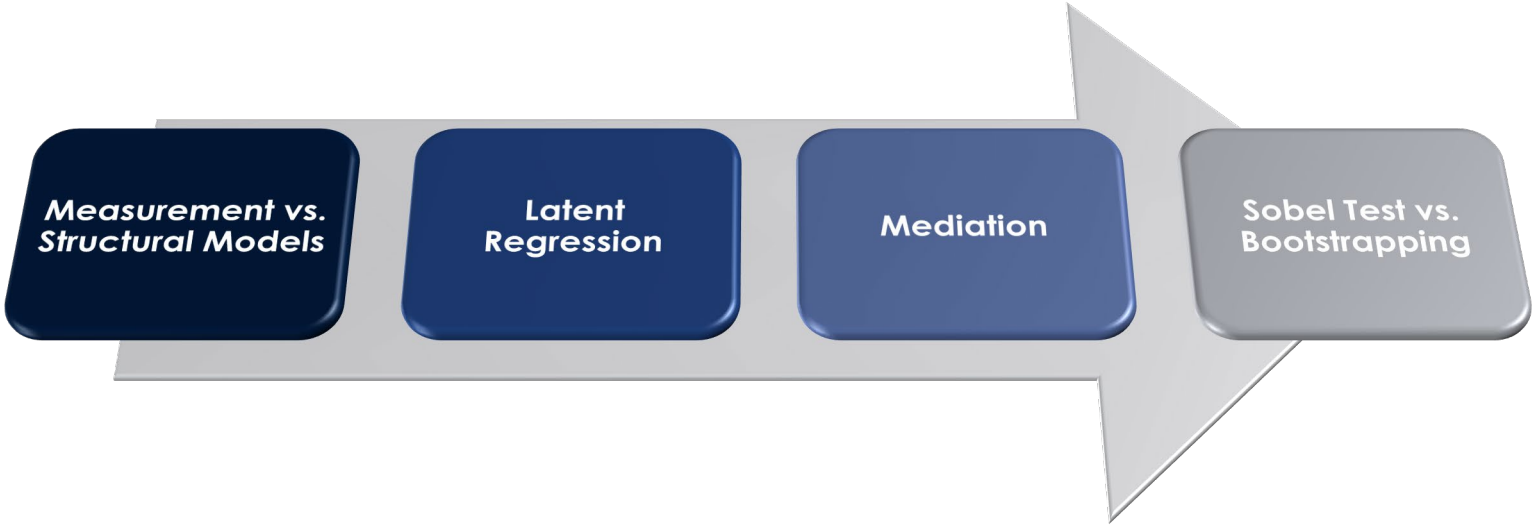
Dept of Educational Psychology, Leadership, & Counseling

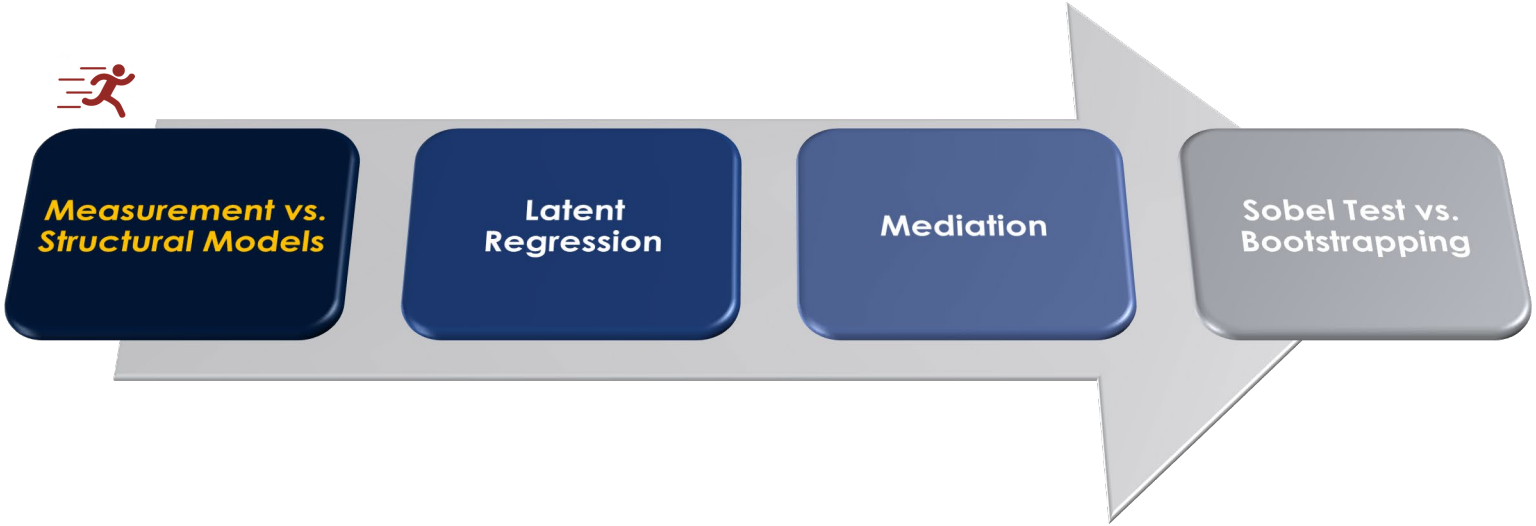
EPSY 5383

DATA ANALYSIS WITH STATISTICAL SOFTWARE

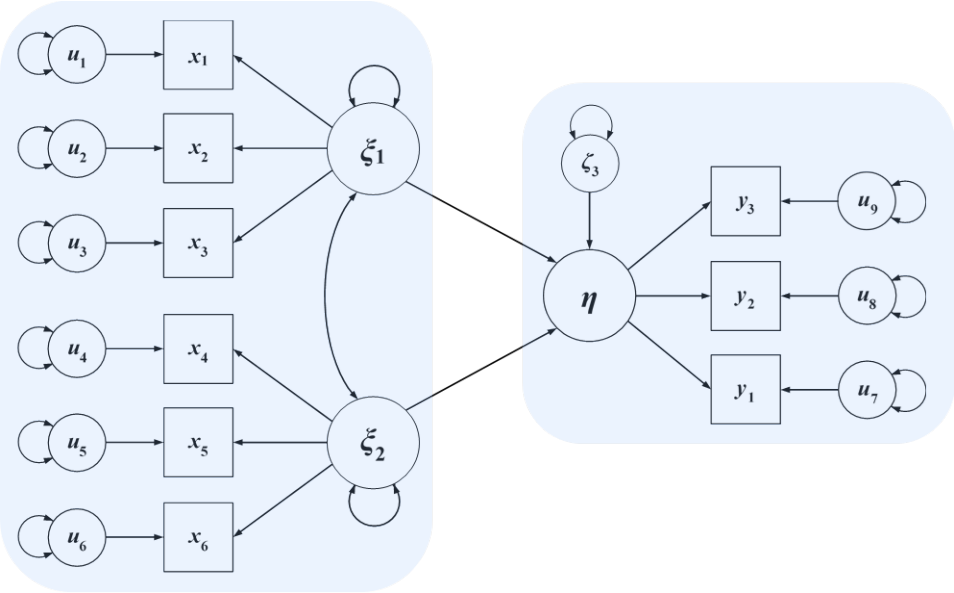
Module Overview



Measurement vs. Structural Models



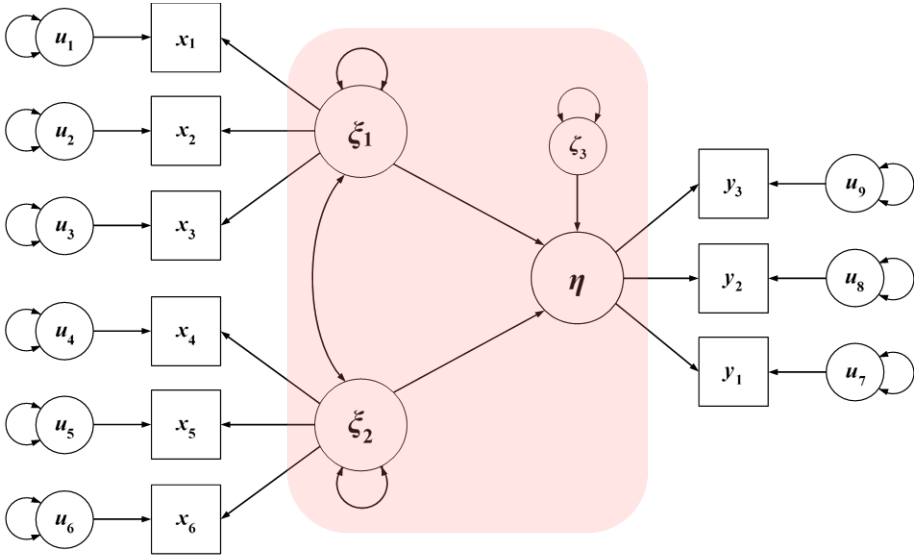
Measurement Model



SEM (structural equation modeling) is composed of two parts:

- ***Measurement model*** "measures" latent constructs or factors by using observed variables (i.e., factor loadings)

Structural Model

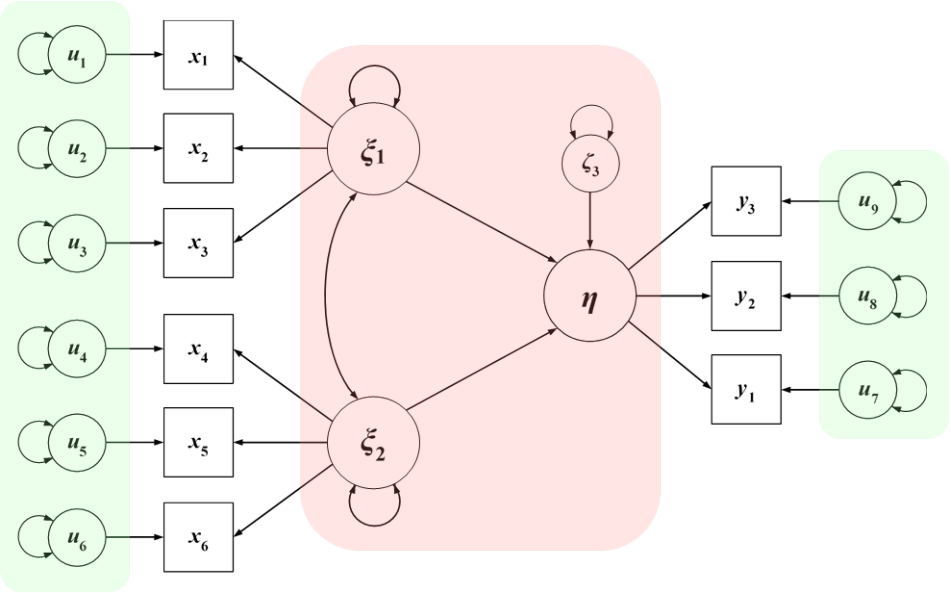


SEM (structural equation modeling) is composed of two parts:

- ***Measurement model*** "measures" latent constructs or factors by using observed variables (i.e., factor loadings)

- ***Structural model*** examines the relationships (i.e., latent regression) among factors.

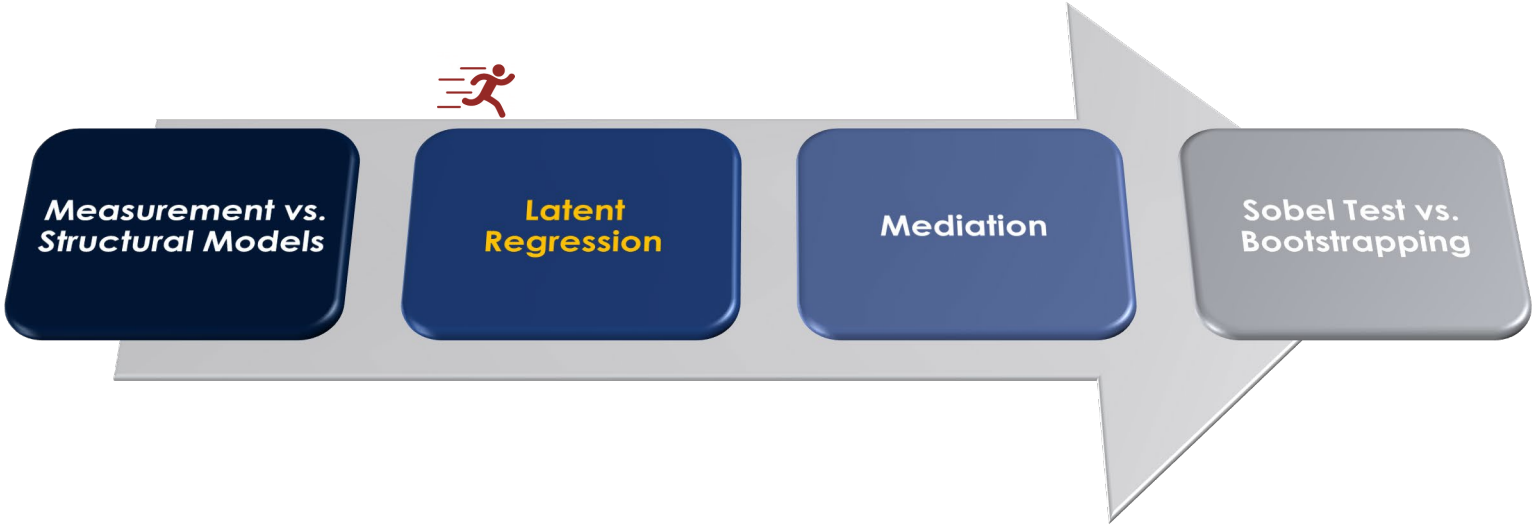
Structural Model



You could run ***OLS regression*** using "composite scores" for the constructs (e.g., average of observed variables), but ***latent regression*** has several advantages over OLS regression:

- Corrects for <u>measurement error</u> (i.e., unreliability of the instruments).
- Can incorporate <u>multiple</u> "dependent" variables (constructs).
- Can specify complex hypotheses (e.g., **mediation**, **moderation**) into a single model.

Latent Regression



Running Example

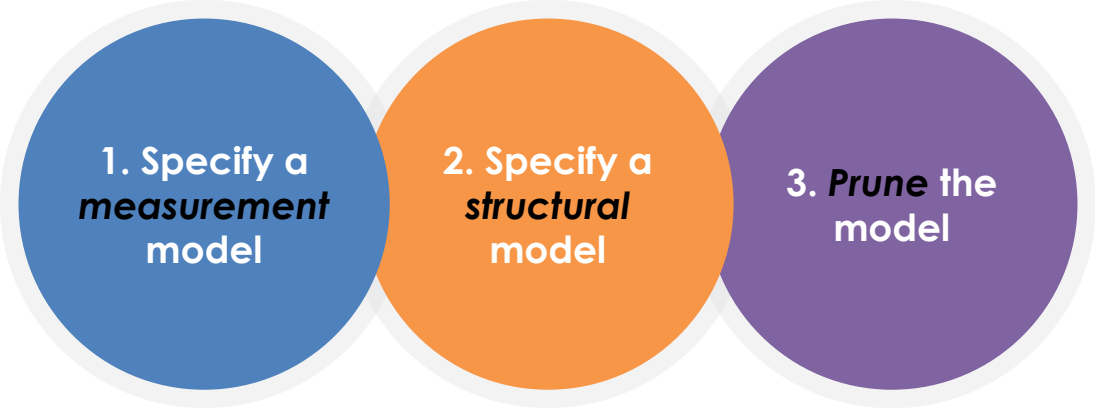
PANAS7 data

- 380 students in 7th grade
- **Positive Affect and Negative Affect Schedule (PANAS)**, a self-report measure of *internalizing symptoms*
- 5 factors, each measured by 3 observed variables
  - *Positive Affect*
  - *Negative Affect*
  - *Agency*
  - *Intrinsic Motivation*
  - *Extrinsic Motivation*

	Very slightly/ Not at all	A little	Moderately	Quite a bit	Extremely
Interested	1	2	3	4	5
Alert	1	2	3	4	5
Attentive	1	2	3	4	5
Excited	1	2	3	4	5
Enthusiastic	1	2	3	4	5
Inspired	1	2	3	4	5
Proud	1	2	3	4	5
Determined	1	2	3	4	5
Strong	1	2	3	4	5
Active	1	2	3	4	5
Upset	1	2	3	4	5
Distressed	1	2	3	4	5

Steps for Latent Regression

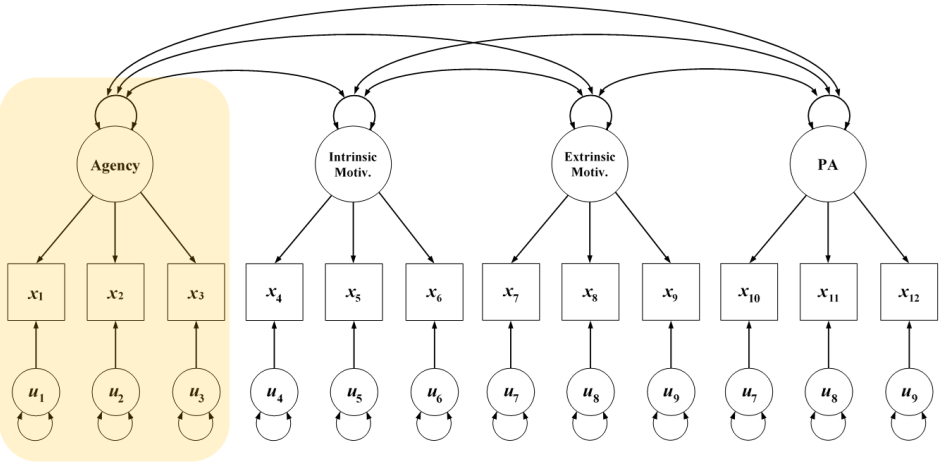
Latent regression usually proceeds in 3 steps:



Measurement Model

<u>Step 1</u>: ***Specify a measurement model*** *(i.e., CFA)*

- This step involves (1) identifying the *factor structure* (item-factor connections) and (2) estimating *parameters* and *fit* of the model.



<u>*R*</u>:

```
m1 ← ' Agency   =~ agency1 + agency2 + agency3
       Intrin   =~ intrin1 + intrin2 + intrin3
       Extrin   =~ extrin1 + extrin2 + extrin3
       PosAff   =~ posaff1 + posaff2 + posaff3 '
```

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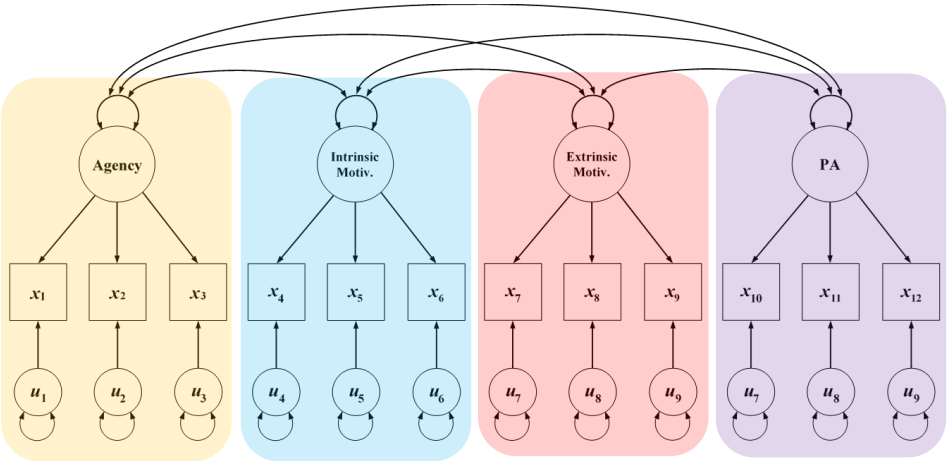

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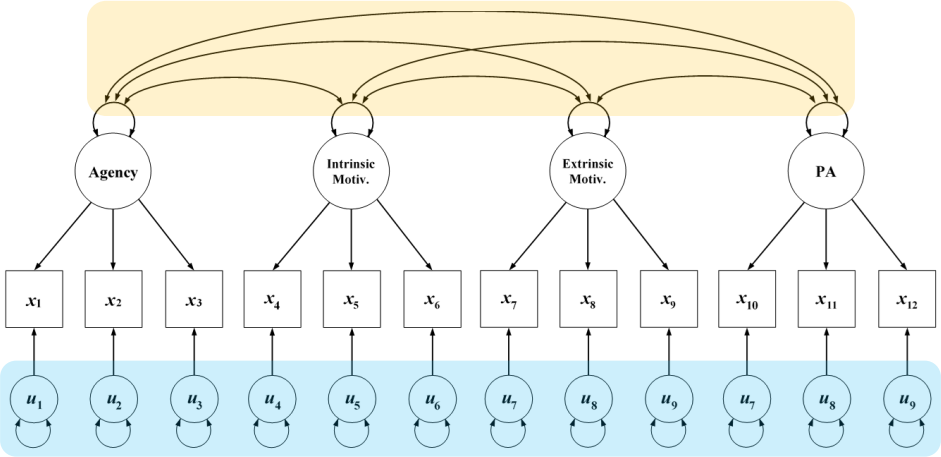

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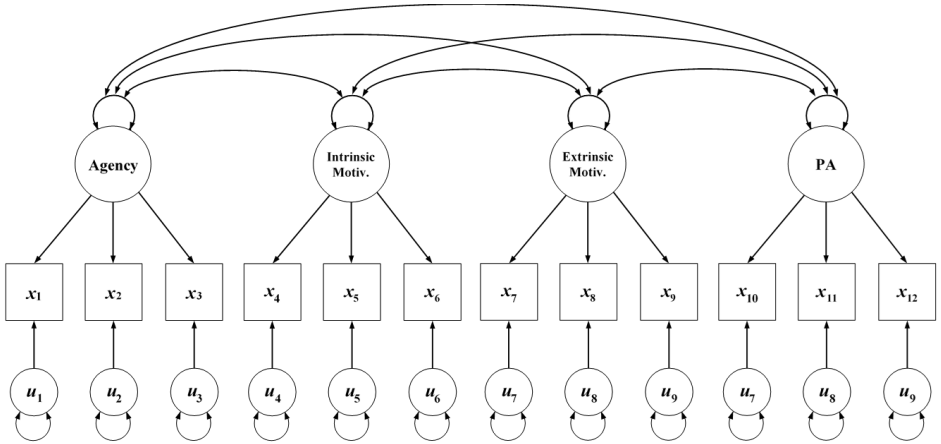
```

- By default,
  - All *factor correlations* are estimated.
  - All *unique factor variances* are estimated.

Measurement Model

Step 1: ***Specify a measurement model*** *(i.e., CFA)*



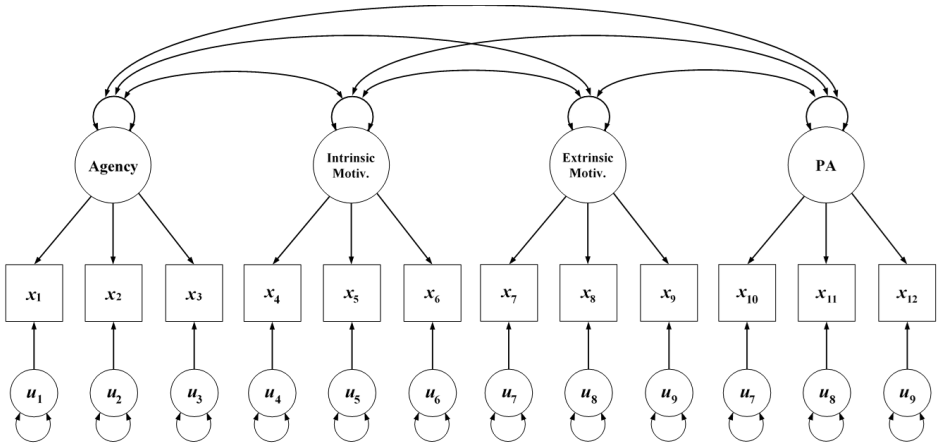
R:

```
out1 <- cfa(model=m1, data=panas7, std.lv=TRUE,
meanstructure=TRUE)
```

- std.lv=TRUE *standardizes* each factor (i.e., *mean* = 0, variance = 1).
- meanstructure=TRUE estimates *intercepts of observed variables*.

Measurement Model

Step 1: ***Specify a measurement model*** *(i.e., CFA)*



R:

```
summary(out1, standardized=TRUE,
fit.measures=TRUE, rsquare=TRUE)
```

- standardized=TRUE provides *standardized* parameter estimates in addition to unstandardized estimates.
- fit.measures=TRUE provides tests and measures of *model fit*.
- rsquare=TRUE provides the *proportion of explained variance* in each observed variable.

```
lavaan 0.6-9 ended normally after 54 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                        42

  Number of observations                           380

Model Test User Model:

  Test statistic                                57.525
  Degrees of freedom                                48
  P-value (Chi-square)                           0.163

Model Test Baseline Model:

  Test statistic                              2806.169
  Degrees of freedom                                66
  P-value                                        0.000

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    0.997
  Tucker-Lewis Index (TLI)                       0.995

Loglikelihood and Information Criteria:

  Loglikelihood user model (H0)              -2976.044
  Loglikelihood unrestricted model (H1)      -2947.282

  Akaike (AIC)                                6036.088
  Bayesian (BIC)                              6201.576
  Sample-size adjusted Bayesian (BIC)         6068.318

Root Mean Square Error of Approximation:

  RMSEA                                          0.023
  90 Percent confidence interval - lower         0.000
  90 Percent confidence interval - upper         0.042
  P-value RMSEA <= 0.05                          0.992

Standardized Root Mean Square Residual:

  SRMR                                           0.026
```

Model

$\chi^2(48) = 57.53$, $p = .16$, CFI = 1.00, NNFI = 1.00, RMSEA = 0.02 [0.00; 0.04]

Overall goodness-of-fit ***(chi-square) test***

CFI, ***TLI*** (NNFI)

- > 0.90 or 0.95 *good fit*

RMSEA

- < 0.05 *good fit*
- 0.05–0.08 reasonable fit
- 0.08–0.10 mediocre fit
- > 0.10 poor fit

del

Latent Variables:

	Estimate	Std.Err	z-value	P(>|z|)	Std.lv	Std.all
Agency =~						
agency1	0.466	0.021	22.559	0.000	0.466	0.905
agency2	0.492	0.021	23.764	0.000	0.492	0.933
agency3	0.497	0.022	22.826	0.000	0.497	0.911
Intrin =~						
intrin1	0.539	0.039	13.918	0.000	0.539	0.702
intrin2	0.579	0.043	13.435	0.000	0.579	0.681
intrin3	0.618	0.038	16.272	0.000	0.618	0.803
Extrin =~						
extrin1	0.386	0.022	17.454	0.000	0.386	0.804
extrin2	0.456	0.027	17.116	0.000	0.456	0.791
extrin3	0.473	0.026	18.204	0.000	0.473	0.831
PosAff =~						
posaff1	0.570	0.028	20.157	0.000	0.570	0.854
posaff2	0.604	0.029	21.175	0.000	0.604	0.883
posaff3	0.631	0.029	21.600	0.000	0.631	0.894

Covariances:

	Estimate	Std.Err	z-value	P(>|z|)	Std.lv	Std.all
Agency ~~						
Intrin	0.507	0.047	10.795	0.000	0.507	0.507
Extrin	0.270	0.054	5.035	0.000	0.270	0.270
PosAff	0.302	0.051	5.947	0.000	0.302	0.302
Intrin ~~						
Extrin	-0.027	0.063	-0.433	0.665	-0.027	-0.027
PosAff	0.413	0.052	7.910	0.000	0.413	0.413
Extrin ~~						
PosAff	-0.029	0.058	-0.494	0.621	-0.029	-0.029

Intercepts:

	Estimate	Std.Err	z-value	P(>|z|)	Std.lv	Std.all
.agency1	2.442	0.026	92.349	0.000	2.442	4.737
.agency2	2.550	0.027	94.360	0.000	2.550	4.841
.agency3	2.544	0.028	90.926	0.000	2.544	4.664
.intrin1	3.002	0.039	76.152	0.000	3.002	3.907
.intrin2	2.987	0.044	68.467	0.000	2.987	3.512
.intrin3	3.080	0.039	78.022	0.000	3.080	4.002
.extrin1	1.615	0.025	65.603	0.000	1.615	3.365
.extrin2	1.686	0.030	57.065	0.000	1.686	2.927
.extrin3	1.633	0.029	55.873	0.000	1.633	2.866
.posaff1	3.136	0.034	91.663	0.000	3.136	4.702
.posaff2	2.991	0.035	85.211	0.000	2.991	4.371
.posaff3	3.069	0.036	84.776	0.000	3.069	4.349
Agency	0.000				0.000	0.000
Intrin	0.000				0.000	0.000
Extrin	0.000				0.000	0.000
PosAff	0.000				0.000	0.000

Standardized ***factor loading*** and *p*-value

- > 0.50 *good item reliability*
- In other words, more than 75% of the variance in the observed variable is explained by the factor

del

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PosAff	0.000				0.000	0.000

Factor correlation and *p*-value

del

```
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                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  Agency =~
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    Agency            0.000                             0.000    0.000
    Intrin            0.000                             0.000    0.000
    Extrin            0.000                             0.000    0.000
    PosAff            0.000                             0.000    0.000
```

Intercept (mean) of observed variable, added by the command **meanstructure**=TRUE

Measurement Model

Variances:

	Estimate	Std.Err	z-value	P(>|z|)	Std.lv	Std.all
.agency1	0.048	0.005	9.650	0.000	0.048	0.181
.agency2	0.036	0.005	7.622	0.000	0.036	0.129
.agency3	0.050	0.005	9.260	0.000	0.050	0.169
.intrin1	0.300	0.029	10.237	0.000	0.300	0.507
.intrin2	0.388	0.036	10.662	0.000	0.388	0.536
.intrin3	0.210	0.029	7.346	0.000	0.210	0.356
.extrin1	0.081	0.009	9.023	0.000	0.081	0.354
.extrin2	0.124	0.013	9.447	0.000	0.124	0.374
.extrin3	0.101	0.013	7.996	0.000	0.101	0.310
.posaff1	0.120	0.012	9.890	0.000	0.120	0.270
.posaff2	0.103	0.012	8.575	0.000	0.103	0.221
.posaff3	0.100	0.013	7.939	0.000	0.100	0.201
Agency	1.000				1.000	1.000
Intrin	1.000				1.000	1.000
Extrin	1.000				1.000	1.000
PosAff	1.000				1.000	1.000

R-Square:

	Estimate
agency1	0.819
agency2	0.871
agency3	0.831
intrin1	0.493
intrin2	0.464
intrin3	0.644
extrin1	0.646
extrin2	0.626
extrin3	0.690
posaff1	0.730
posaff2	0.779
posaff3	0.799

Unique factor variance

Measurement Model

Variances:

	Estimate	Std.Err	z-value	P(>|z|)	Std.lv	Std.all
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.intrin3	0.210	0.029	7.346	0.000	0.210	0.356
.extrin1	0.081	0.009	9.023	0.000	0.081	0.354
.extrin2	0.124	0.013	9.447	0.000	0.124	0.374
.extrin3	0.101	0.013	7.996	0.000	0.101	0.310
.posaff1	0.120	0.012	9.890	0.000	0.120	0.270
.posaff2	0.103	0.012	8.575	0.000	0.103	0.221
.posaff3	0.100	0.013	7.939	0.000	0.100	0.201
Agency	1.000				1.000	1.000
Intrin	1.000				1.000	1.000
Extrin	1.000				1.000	1.000
PosAff	1.000				1.000	1.000

R-Square:

	Estimate
agency1	0.819
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extrin3	0.690
posaff1	0.730
posaff2	0.779
posaff3	0.799

Standardized ***unique factor variance*** = the proportion of "unexplained" variance in the observed variable

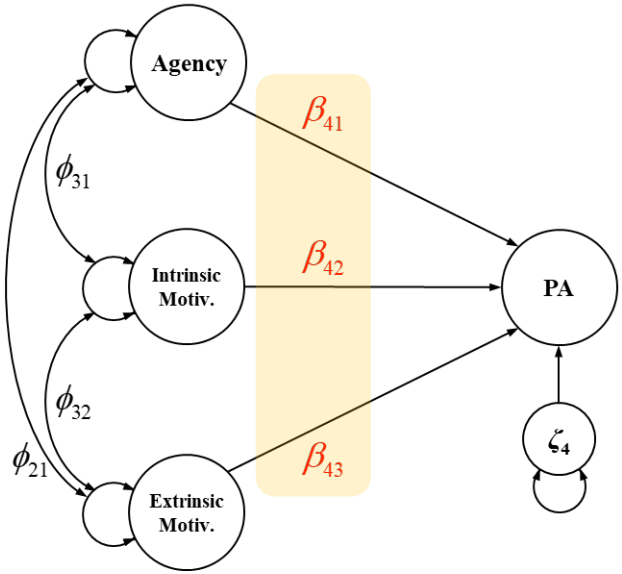
R-square = the proportion of "explained" variance in the observed variable

Standardized ***unique factor variance*** + ***R-square*** = 1 (100%)

Structural Model

<u>Step 2</u>: ***Specify a structural model*** *(regression paths)*

- Factor correlations are converted to *directional (regression) paths* among the factors.



<u>*R*</u>:

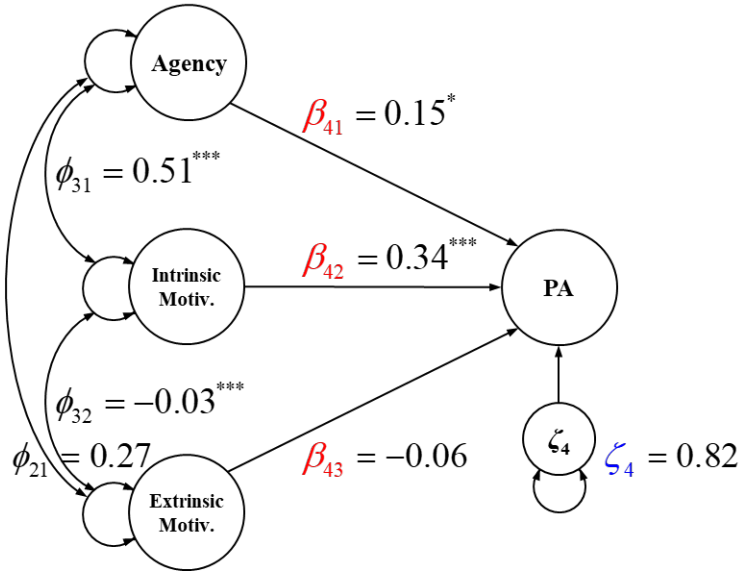
```
m2 ← ‘ Agency    =~ agency1 + agency2 + agency3
       Intrin    =~ intrin1 + intrin2 + intrin3
       Extrin    =~ extrin1 + extrin2 + extrin3
       PosAff    =~ posaff1 + posaff2 + posaff3
       PosAff ~ Agency + Intrin + Extrin ’
out2 ← sem(model=m2, data=panas7, std.lv=TRUE,
meanstructure=TRUE)
```

<u>*Note*</u>. The measurement model is not shown in the figure for simplicity.

Structural Model

<u>Step 2</u>: ***Specify a structural model*** *(regression paths)*

- *Standardized* solution:



$\chi^2(48) = 57.53$, $p = .16$, CFI = 1.00, NNFI = 1.00, RMSEA = 0.02 [0.00; 0.04]

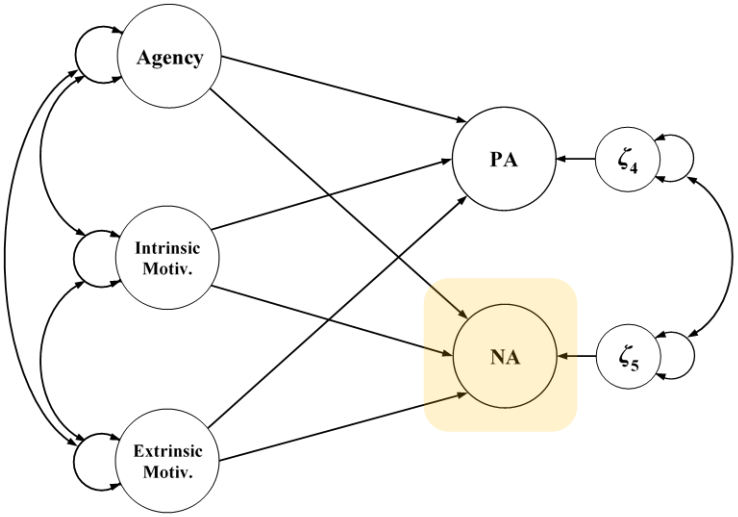
$$R^2 = 1 - \zeta_4 = 1 - 0.82 = 0.18$$

- 18% of the total variance in *Positive Affect* is explained by *Agency*, *Intrinsic Motivation*, and *Extrinsic Motivation*.

Structural Model

<u>Step 2</u>: ***Specify a structural model*** *(regression paths)*

- One of the advantages of using latent regression is the ability to incorporate <u>multiple</u> "dependent" variables (factors). If *Negative Affect* is added to the model,



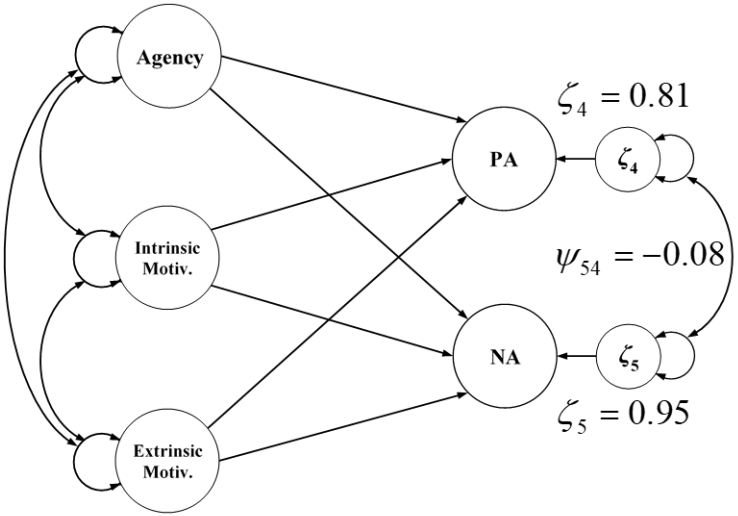
<u>R</u>:

```
m3 ← ' Agency    =~agency1 + agency2 + agency3
       Intrin    =~ intrin1 + intrin2 + intrin3
       Extrin    =~ extrin1 + extrin2 + extrin3
       PosAff    =~ posaff1 + posaff2 + posaff3
       NegAff    =~ negaff1 + negaff2 + negaff3
       PosAff ~ Agency + Intrin + Extrin
       NegAff ~ Agency + Intrin + Extrin '
```

Structural Model

<u>Step 2</u>: ***Specify a structural model*** *(regression paths)*

- One of the advantages of using latent regression is the ability to incorporate <u>multiple</u> "dependent" variables (factors). If *Negative Affect* is added to the model,



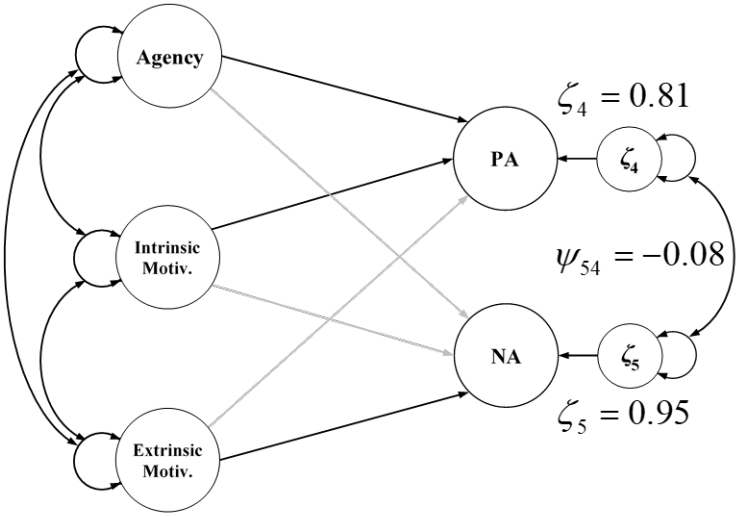
$\chi^2(80) = 106.85$, $p < .05$, CFI = 0.99, NNFI = 0.99, RMSEA = 0.03 [0.01; 0.04]

$$\mathbf{B} = \begin{array}{c} \\ \text{PA} \\ \text{NA} \end{array} \begin{array}{c} \begin{array}{ccc} \text{Agc.} & \text{Intrin.} & \text{Extrin.} \end{array} \\ \begin{bmatrix} 0.15^{*} & 0.34^{***} & -0.06 \\ -0.08 & 0.07 & 0.23^{***} \end{bmatrix} \end{array}$$

Final Model

Step 3: ***Prune the model***

- Non-significant paths are also meaningful outcomes by themselves, but you can "prune" them and have a cleaner (more parsimonious) model.



$\chi^2(80) = 106.85$, $p < .05$, CFI = 0.99, NNFI = 0.99, RMSEA = 0.03 [0.01; 0.04]

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Final Model

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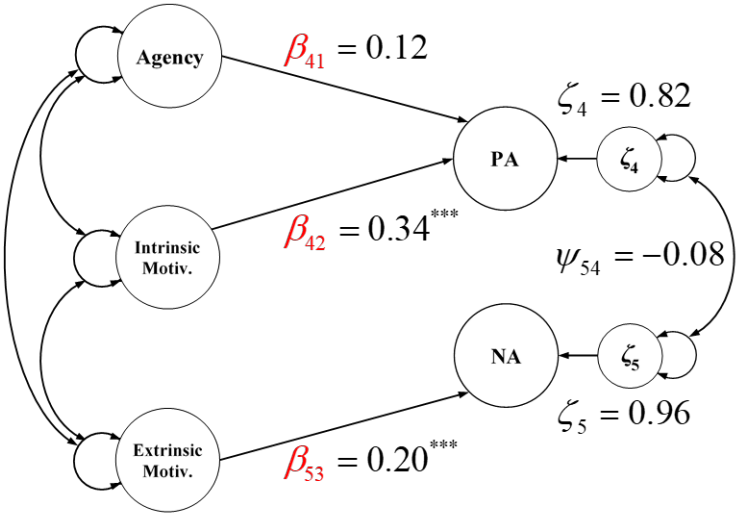
- Non-significant paths are also meaningful outcomes by themselves, but you can "prune" them and have a cleaner (more parsimonious) model.



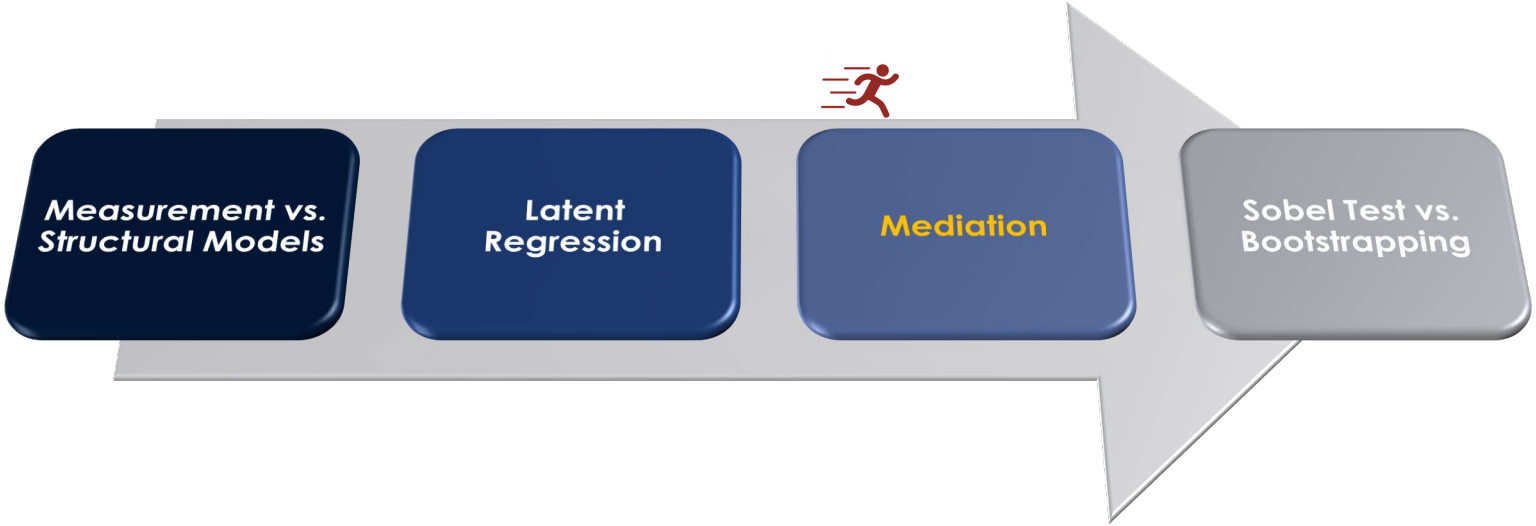
$\chi^2(83) = 109.15$, $p < .05$, CFI = 0.99, NNFI = 0.99, RMSEA = 0.03 [0.01; 0.04]

$\Delta\chi^2(3) = 2.31$, $p = .51$

- LRT results are not significant—i.e., the fit of the pruned model is not significantly worse than the fit of the original model. Thus, accept the pruned model.

18% of the total variance in *Positive Affect* and 4% in *Negative Affect* are explained by *Agency*, *Intrinsic Motivation*, and *Extrinsic Motivation*.

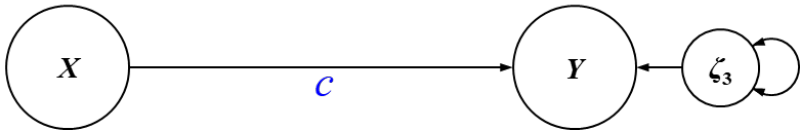
Mediation


Measurement vs. Structural Models
Latent Regression
Mediation
Sobel Test vs. Bootstrapping

Mediation

Let's assume that ***X*** causes ***Y***. In other words, changes in ***X*** make changes in ***Y***.

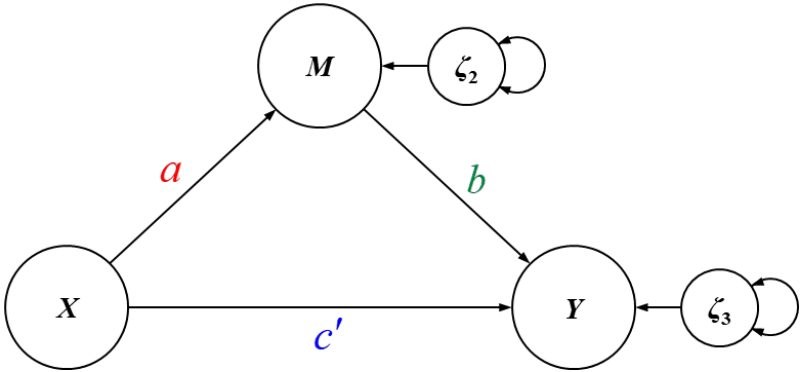
- The (regression) path *c* is called ***total effect*** of ***X*** on ***Y***.



Mediation

A part of the influence of ***X*** on ***Y*** may be "delivered" through ***M***. In other words, changes in ***X*** make changes in ***Y*** directly, as well as indirectly by making changes in ***M*** (***mediator***).

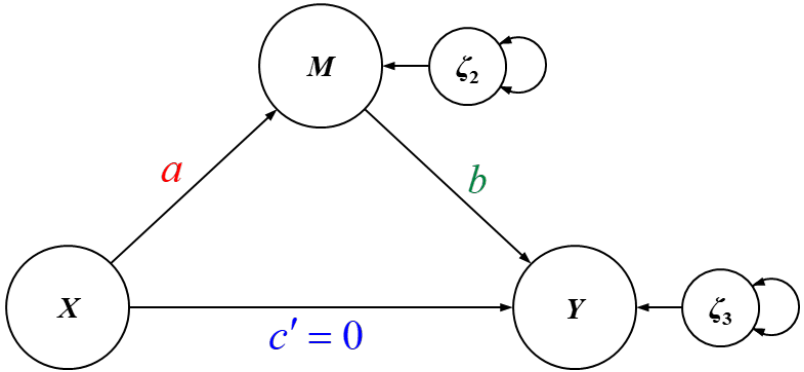
- Now, the path c' is called ***direct effect*** of ***X*** on ***Y***.



Mediation

There are two types of mediation: Full and partial.

- **Full mediation** occurs when ***X*** no longer affects ***Y***, after ***M*** has been introduced. All the influence of ***X*** on ***Y*** is "fully" mediated by ***M***.
    - In such case, you can remove the path c' from the model—i.e., "casual chain" (***X***-***M***-***Y***).



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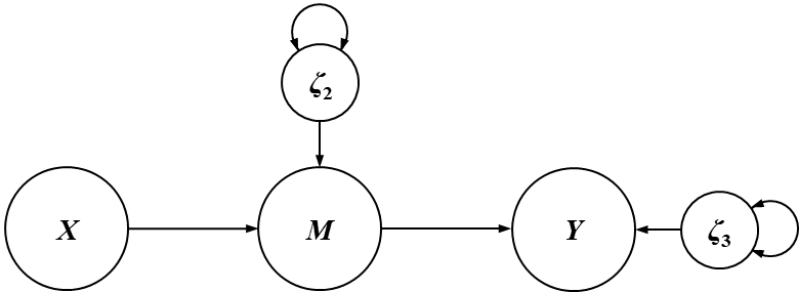
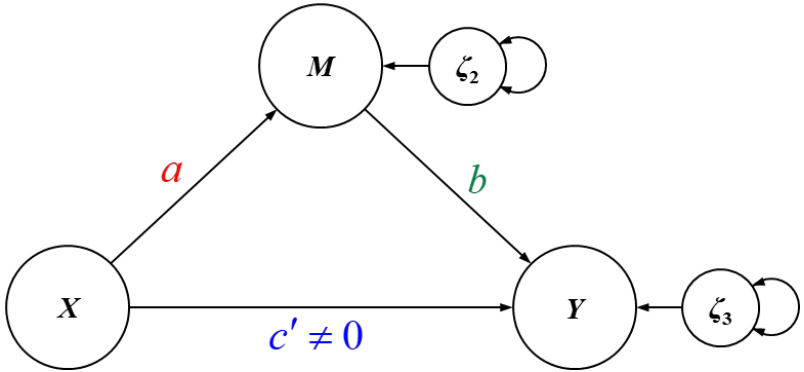


Mediation

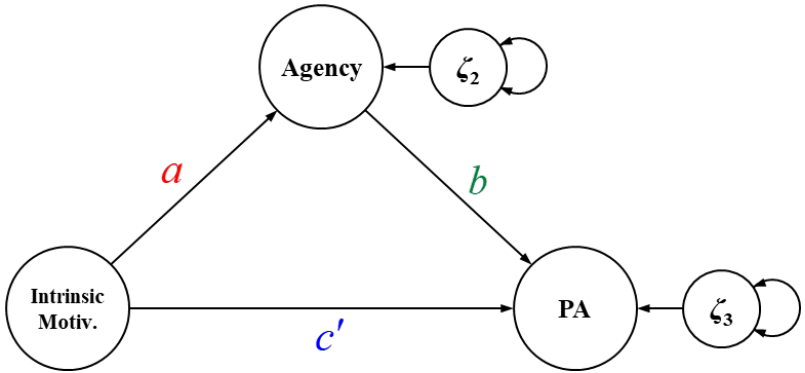
There are two types of mediation: Full and partial.

- **Partial mediation** occurs when the influence of ***X*** is reduced (in size) but is still different from 0 after ***M*** has been introduced. The influence of ***X*** on ***Y*** is "partially" mediated by ***M***.



Steps for Mediation Analysis

To establish mediation, 3 conditions (steps) should be met. (Kenny et al., 1998; Fazier et al., 2004)



<u>*Step 1*</u>: Show that ***X*** affects ***M***.

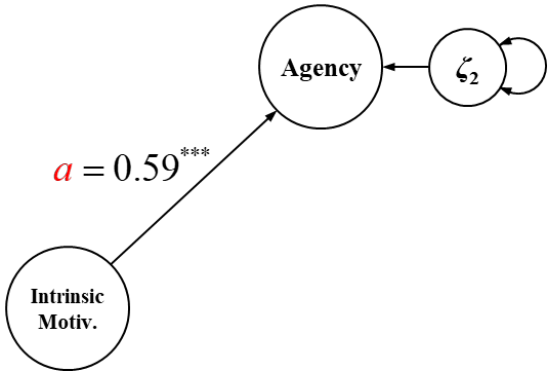
<u>*Step 2*</u>: Show that ***M*** affects ***Y*** when controlling for ***X***.

<u>*Step 3*</u>: Determine whether mediation is full or partial.

Steps for Mediation Analysis

To establish mediation, 3 conditions (steps) should be met.

$\chi^2(8) = 10.71$, $p = .22$
CFI = 1.00, NNFI = 1.00,
RMSEA = .03 [.00; .07]



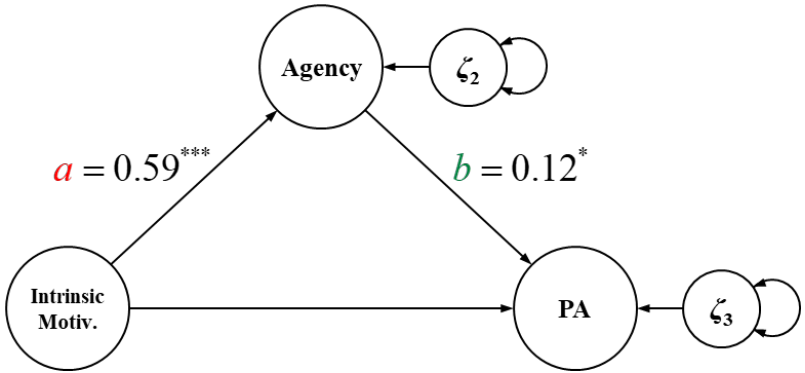
✓ <u>*Step 1*</u>: Show that ***X*** affects ***M***.

- *Intrinsic Motivation* significantly predicts *Agency*.

Steps for Mediation Analysis

To establish mediation, 3 conditions (steps) should be met.

$\chi^2(24) = 22.38$, $p = .56$
CFI = 1.00, NNFI = 1.00,
RMSEA = .00 [.00; .04]



✓ <u>*Step 1*</u>: Show that ***X*** affects ***M***.

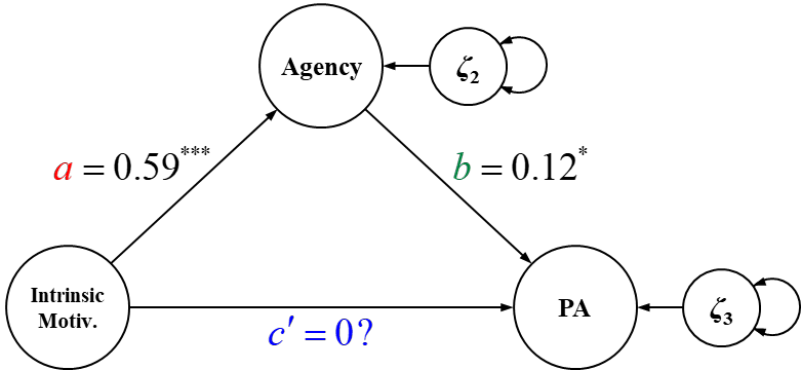
✓ <u>*Step 2*</u>: Show that ***M*** affects ***Y*** when controlling for ***X***.

- *Agency* significantly predicts *Positive Affect*, when controlling for *Intrinsic Motivation*.

Steps for Mediation Analysis

To establish mediation, 3 conditions (steps) should be met.

$\chi^2(24) = 22.38$, $p = .56$
CFI = 1.00, NNFI = 1.00,
RMSEA = .00 [.00; .04]



✓ *Step 1*: Show that ***X*** affects ***M***.

✓ *Step 2*: Show that ***M*** affects ***Y*** when controlling for ***X***.

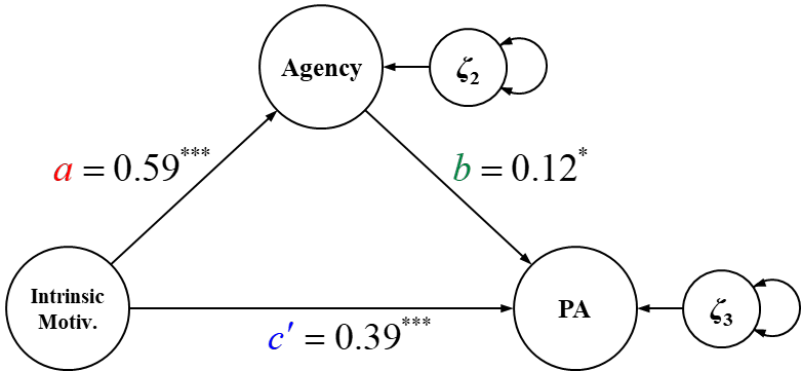
Step 3: Determine whether mediation is full or partial.

- If $c' = 0$, full mediation.
- Otherwise, partial mediation.

Steps for Mediation Analysis

To establish mediation, 3 conditions (steps) should be met.

$\chi^2(24) = 22.38$, $p = .56$
CFI = 1.00, NNFI = 1.00,
RMSEA = .00 [.00; .04]



✓ *Step 1*: Show that ***X*** affects ***M***.

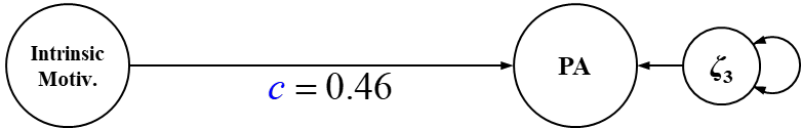
✓ *Step 2*: Show that ***M*** affects ***Y*** when controlling for ***X***.

✓ *Step 3*: Determine whether mediation is full or partial.

- The direct effect of *Intrinsic Motivation* on *Positive Affect* is significantly different from 0. Thus, partial mediation.

Indirect Effect

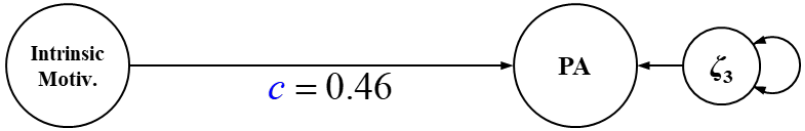
Once mediation is supported, the additional final step is to measure the amount of mediation.



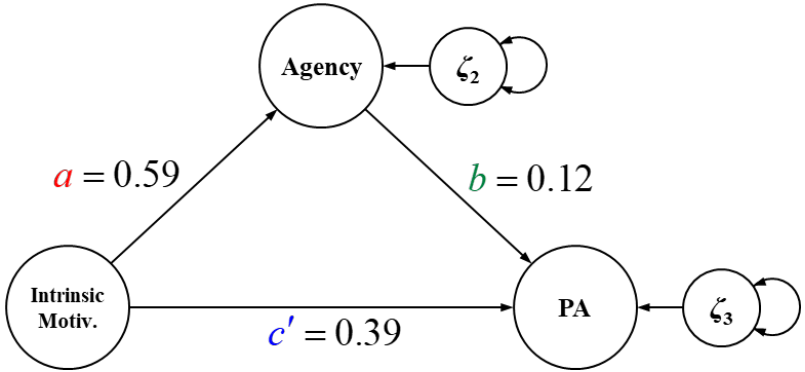
Recall that the path c represents the ***total effect*** of *Intrinsic Motivation* on *Positive Affect*.

Indirect Effect

Once mediation is supported, the additional final step is to measure the amount of mediation.



Recall that the path c represents the ***total effect*** of *Intrinsic Motivation* on *Positive Affect*.



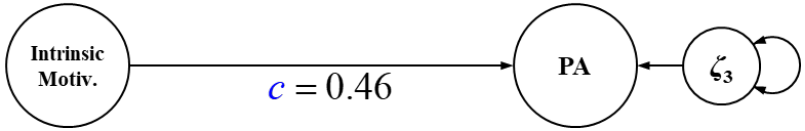
This total effect is partitioned into the ***direct effect*** (c'), and the ***indirect effect*** through *Agency*.

Thus, the amount of mediation equals to the amount of **indirect effect**, the product of a and b:

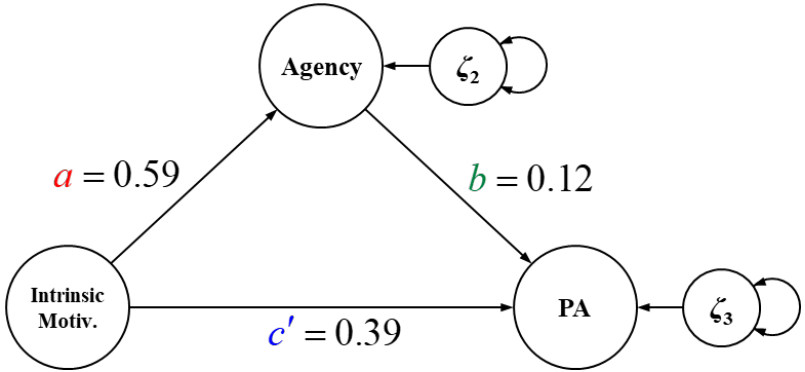
$$c = c' + ab$$

Indirect Effect

Once mediation is supported, the additional final step is to measure the amount of mediation.



Recall that the path c represents the ***total effect*** of *Intrinsic Motivation* on *Positive Affect*.



If *Intrinsic Motivation* increases by 1 unit (point), this change raises *Positive Affect* by 0.39 and *Agency* by 0.59.

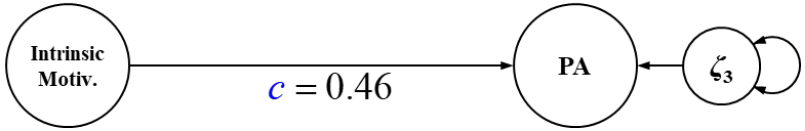
The 0.59 increase in *Agency* also raises *Positive Affect* by (0.59 × 0.12 =) 0.07.

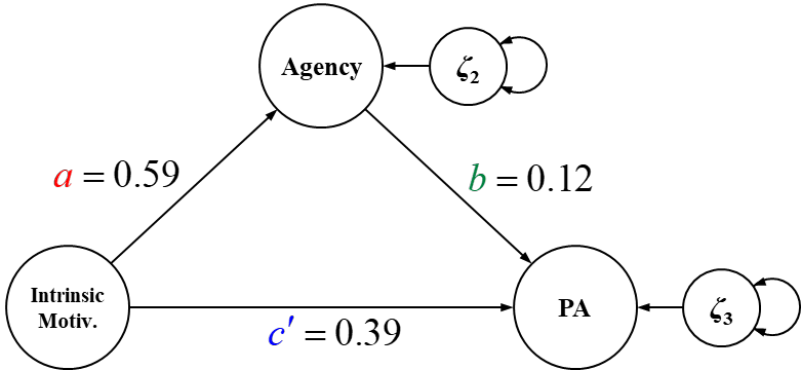
$$c = c' + ab$$

$$0.46 = 0.39 + (0.59)(0.12)$$

Indirect Effect

Once mediation is supported, the additional final step is to measure the amount of mediation.



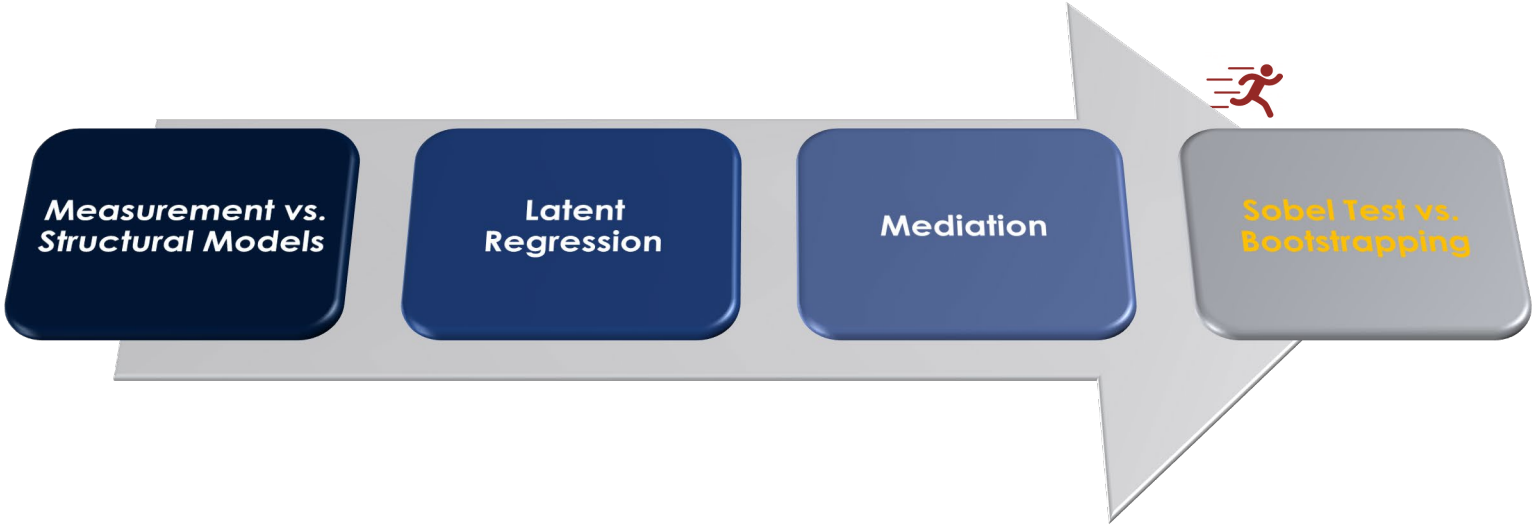


Then, how can you decide statistical significance of the product term ab (indirect effect)?

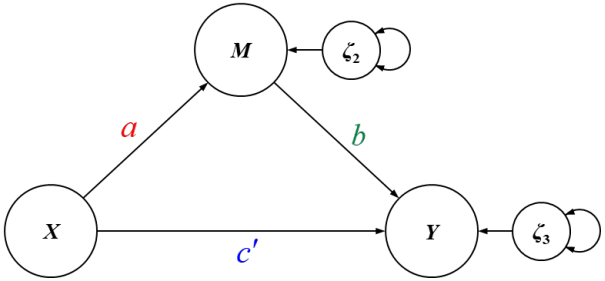
A few different approaches have been proposed in the literature. The most widely used methods are:

- Sobel test
- Bootstrapping

Sobel Test vs. Bootstrapping



Sobel Test

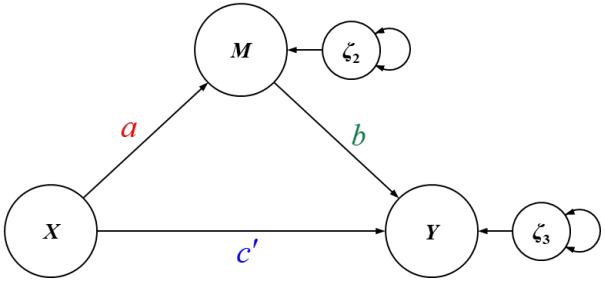


Sobel test (Sobel, 1982) provides an *approximate* estimate of ***standard error*** of *ab* and test statistic *z*:

$$SE_{ab} = \sqrt{a^2 s_b^2 + b^2 s_a^2} \qquad z = \frac{ab}{SE_{ab}}$$

- If |*z*| > 1.645, the indirect effect is significant at .10 alpha level (i.e., $p < .10$).
- If |*z*| > 1.960, the indirect effect is significant at .05 alpha level (i.e., $p < .05$).
- If |*z*| > 2.580, the indirect effect is significant at .01 alpha level (i.e., $p < .01$).

Sobel Test



$$SE_{ab} = \sqrt{a^2 s_b^2 + b^2 s_a^2}$$
$$= \sqrt{\left(0.59^2\right)\left(0.06^2\right) + \left(0.12^2\right)\left(0.07^2\right)}$$
$$= 0.04$$

$$z = \frac{ab}{SE_{ab}} = \frac{(0.59)(0.12)}{0.04} = 1.90$$

- This *z*-value is less than 1.96. Thus, the indirect effect (i.e., amount of mediation) is <u>not</u> significantly different from 0 at .05 alpha level.

Sobel Test

Limitations:

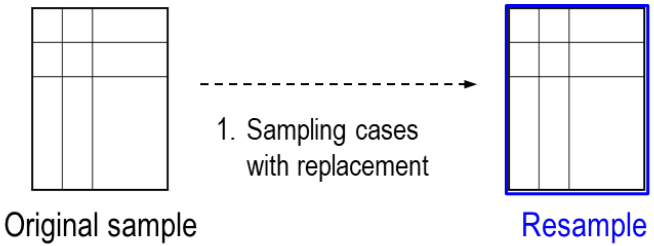
- Sobel test assumes that the estimates of *a* and *b* are independent, which is not always true.
- Because Sobel test uses a normal approximation (i.e., *z* test), it requires a normal distribution of *ab*. However, the sampling distribution of *ab* is highly skewed.
- Consequently, Sobel test is *very conservative* (hard to reject the null hypothesis that *ab* = 0, as we just observed in our example) and has *very low power*. (MacKinnon et al., 1995)
- Thus, the use of Sobel test should be avoided.

Bootstrapping

Bootstrapping (Shrout & Bolger, 2002) is an increasingly popular non-parametric test for indirect effect.

Bootstrapping

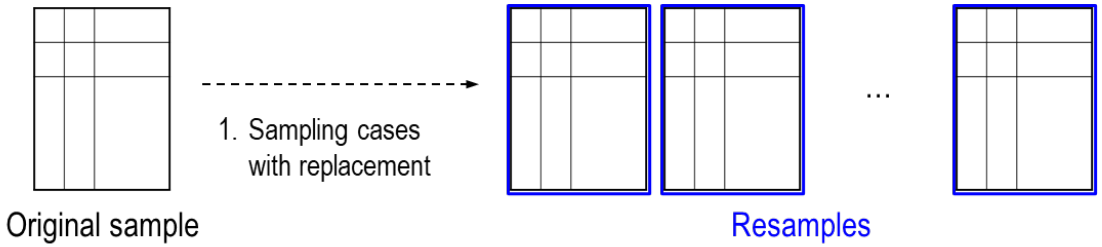
Step 1: *N* cases are drawn (*with replacement*) from the original sample of *N* cases. A case can be drawn more than once into this "resample".



Bootstrapping

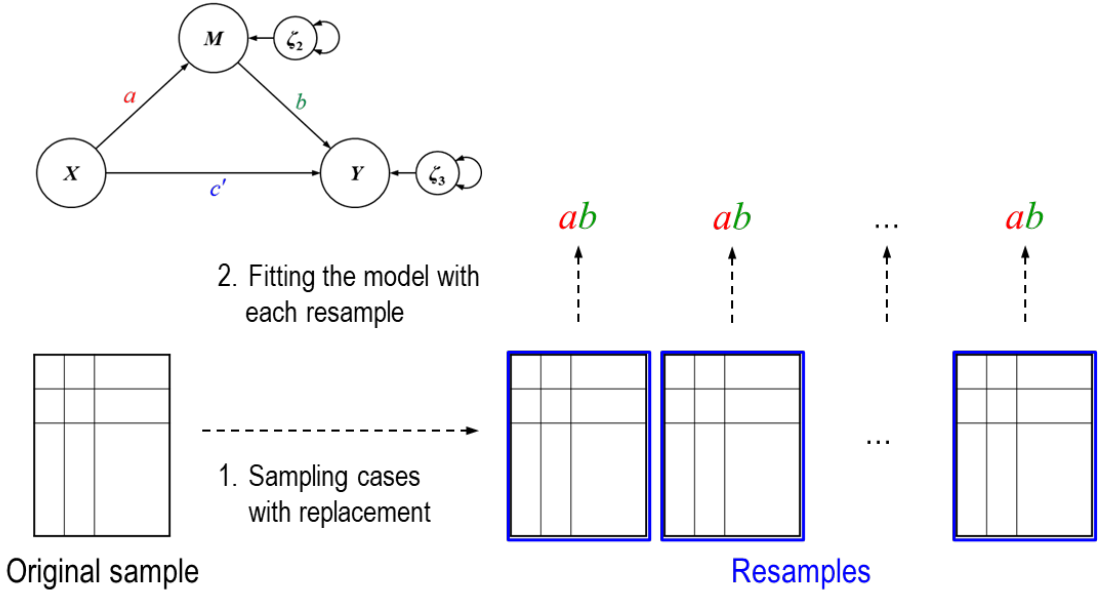
<u>*Step 1*</u>: *N* cases are drawn (*with replacement*) from the original sample of *N* cases. A case can be drawn more than once into this "resample".

- Create many (e.g., 5,000) resamples.



Bootstrapping

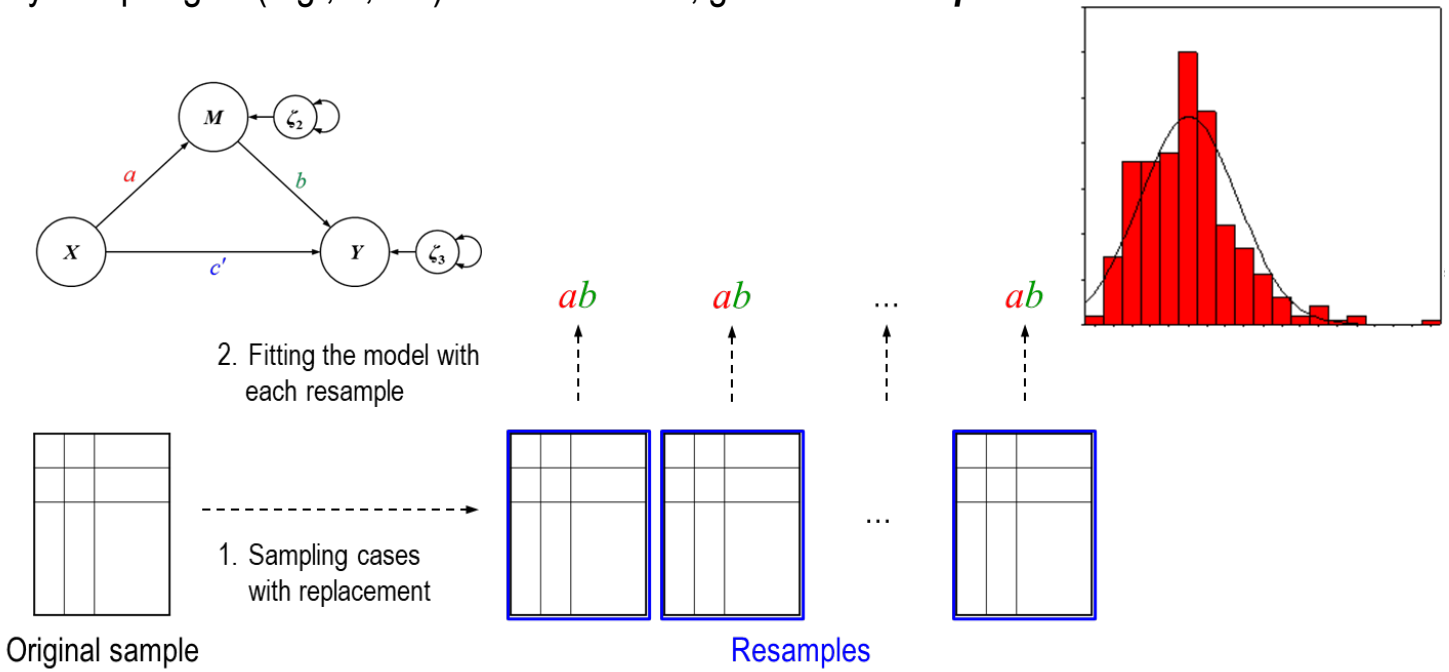
Step 2: Estimate *ab* with each resample.



Bootstrapping

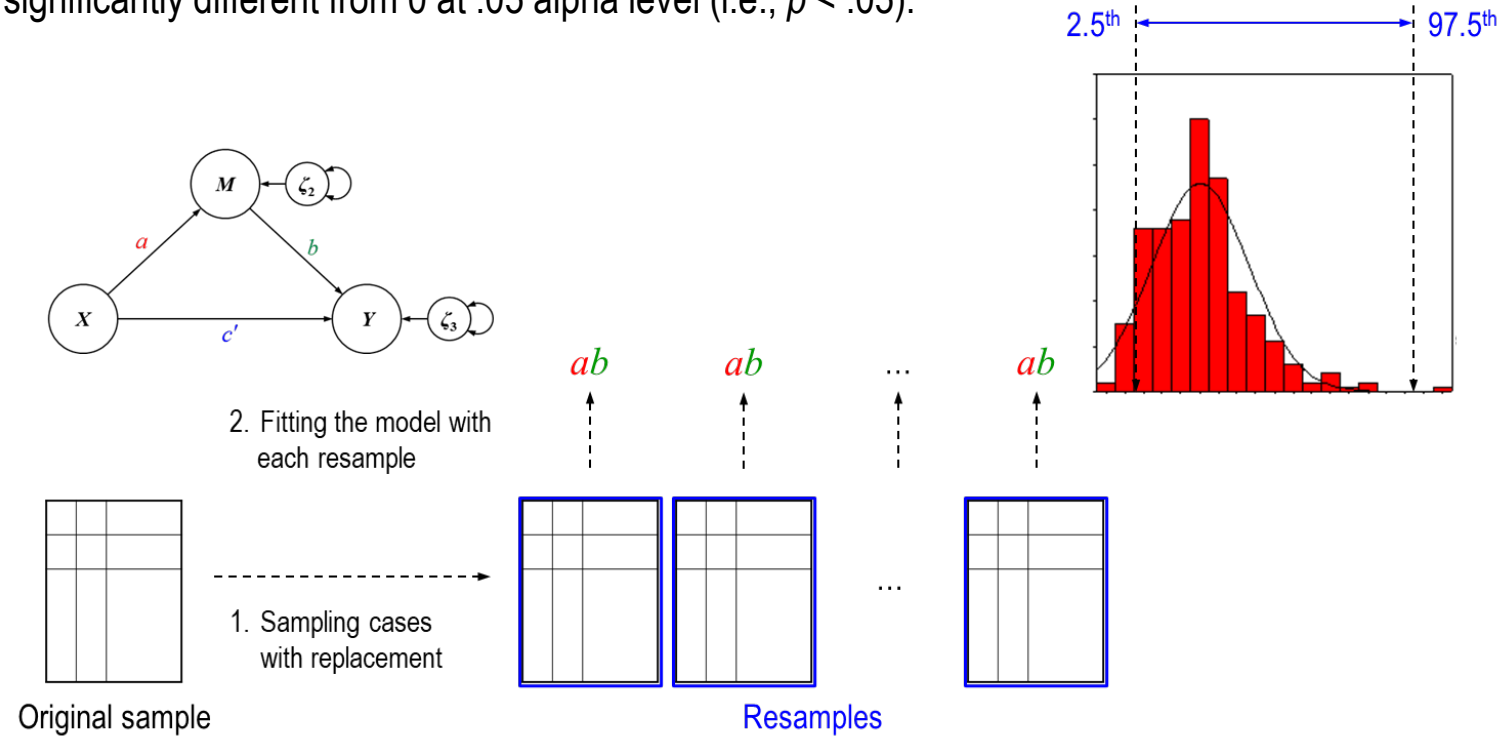
<u>*Step 2*</u>: Estimate *ab* with each resample.

- By compiling all (e.g., 5,000) estimates of *ab*, generate an ***empirical distribution*** of *ab*.



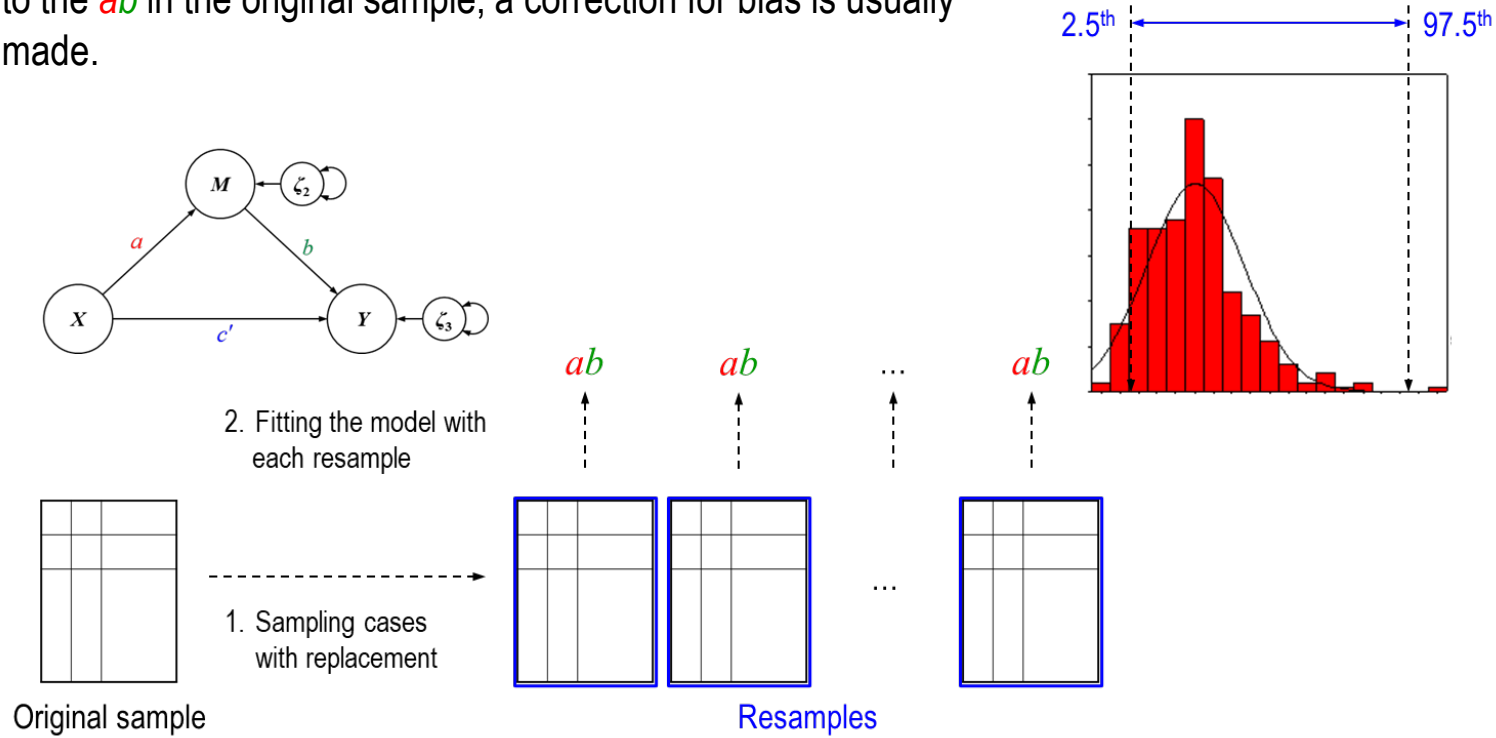
Bootstrapping

Step 3. Find a confidence interval (CI) from the empirical distribution of *ab*. For instance, if a 95% CI does not include 0, *ab* is significantly different from 0 at .05 alpha level (i.e., $p < .05$).

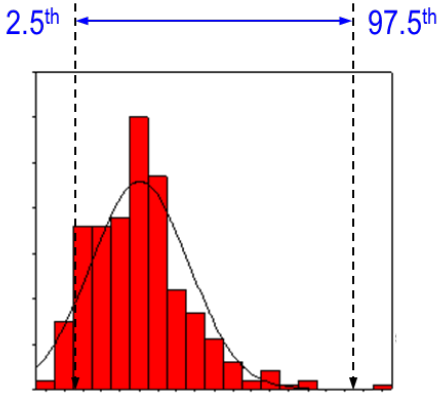


Bootstrapping

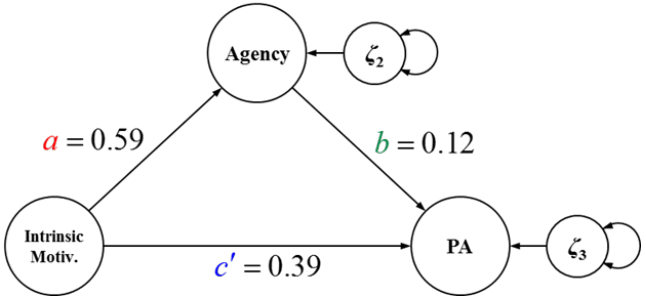
- When the mean of all (e.g., 5,000) estimates of *ab* is <u>not</u> close to the *ab* in the original sample, a correction for bias is usually made.



Bootstrapping



$$ab = 0.07[0.002, 0.15]$$



R:

```
…
Agency ~ a*Intrin
Positive ~ b*Agency
Positive ~ c*Intrin

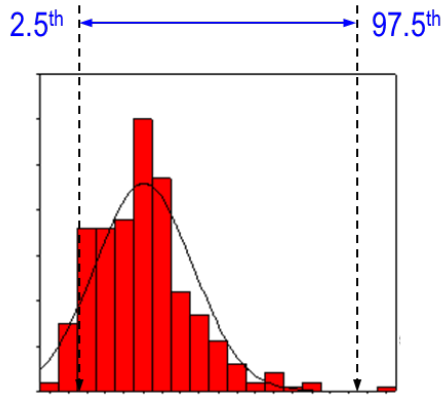
ab := a * b
total := c + (a * b) '

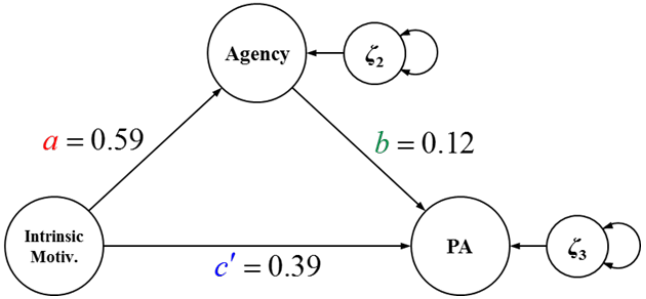
out8 ← sem(m8, data=panas7, std.lv=TRUE, meanstructure=TRUE,
se="bootstrap", bootstrap=2000)

parameterEstimates(out8, level=0.95, boot.ci.type="bca.simple")
```

- The bca.simple option provides bias-adjusted bootstrap confidence intervals.

Bootstrapping



lhs <chr>	op <chr>	rhs <chr>	label <chr>	est <dbl>	se <dbl>	z <dbl>	pvalue <dbl>	ci.lower <dbl>	ci.upper <dbl>
posaff1	~1			3.136	0.035	90.455	0.000	3.065	3.201
posaff2	~1			2.991	0.035	84.705	0.000	2.922	3.060
posaff3	~1			3.069	0.037	83.881	0.000	2.997	3.140
Agency	~1			0.000	0.000	NA	NA	0.000	0.000
Intrin	~1			0.000	0.000	NA	NA	0.000	0.000
PosAff	~1			0.000	0.000	NA	NA	0.000	0.000
ab	:=	a*b	ab	0.070	0.039	1.787	0.074	0.003	0.157
total	:=	c+(a*b)	total	0.456	0.070	6.504	0.000	0.327	0.601

$$ab = 0.07[0.002, 0.15]$$